

## AD 502 – Machine Learning

### Pre-Reading Material – Lecture 29

#### Application of Clustering

## PRE-READING: APPLICATIONS OF CLUSTERING

### Introduction

**Clustering** is an unsupervised machine learning technique used to group similar data objects into meaningful groups called **clusters**.

Unlike classification, clustering does not require predefined class labels. The algorithm identifies natural patterns or groups within the data based on similarities among objects.

### Example:

A retail company can group customers into different segments based on purchasing behavior, such as:

- Frequent buyers
- Occasional buyers
- High-value customers
- Price-sensitive customers

### Why is Clustering Used?

Clustering is useful when the categories or groups in the data are **not known in advance**.

Major purposes of clustering include:

- Discovering hidden patterns
- Grouping similar objects
- Identifying customer segments
- Detecting unusual or abnormal data
- Summarizing large datasets
- Supporting decision-making
- Exploring datasets before applying other machine learning techniques



## Major Applications of Clustering

### A. Customer Segmentation

Businesses use clustering to divide customers into groups based on characteristics such as:

- Purchasing behavior
- Frequency of purchases
- Product preferences
- Spending patterns
- Website activity

#### Example:

An online shopping company may identify groups such as **high-value customers, regular customers, and occasional customers.**

### B. Image Segmentation

Clustering can divide an image into regions containing similar pixels.

Applications include:

- Object detection
- Medical image analysis
- Satellite image analysis
- Face detection
- Image compression

#### Example:

In a medical image, clustering can help separate regions representing different types of tissues.

### C. Document and Text Clustering

Large collections of documents can be grouped according to their content.

For example, news articles can automatically be grouped into:

- Sports
- Politics
- Business
- Technology
- Entertainment

This is useful for **search engines, news websites, and document management systems.**

#### ***D. Medical and Healthcare Applications***

Clustering can help identify groups of patients with similar characteristics.

Applications include:

- Patient segmentation
- Disease pattern identification
- Medical image analysis
- Grouping patients according to symptoms
- Identifying similar treatment-response patterns

For example, patients having similar clinical characteristics can be grouped for further analysis.

#### ***E. Anomaly and Fraud Detection***

Clustering can help identify data points that do not belong to any major group.

Applications include:

- Credit card fraud detection
- Network intrusion detection
- Unusual banking transactions
- Abnormal system behavior

A transaction that is very different from the normal transaction clusters may require further investigation.

### ***F. Social Network Analysis***

Clustering can identify groups of users who have similar connections or interaction patterns.

For example, a social network may contain communities related to:

- Sports
- Education
- Music
- Technology
- Entertainment

This is also known as **community detection** in network analysis.

### ***G. Geographic and Satellite Data***

Clustering can group locations having similar geographical or environmental characteristics.

Applications include:

- Land-use classification
- Weather pattern analysis
- Agricultural analysis
- Urban planning
- Remote sensing
- Disaster management

### ***H. Recommendation Systems***

Clustering can group users or products according to their similarities.

For example:

## **Users with similar preferences → Similar user group → Relevant product recommendations**

This can help recommendation systems provide more personalized suggestions.

### **Clustering in Different Domains**

| <b>Domain</b> | <b>Example Application</b> |
|---------------|----------------------------|
|---------------|----------------------------|

|           |                       |
|-----------|-----------------------|
| Marketing | Customer segmentation |
|-----------|-----------------------|

|            |                  |
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| Healthcare | Patient grouping |
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| Finance | Fraud detection |
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|           |                              |
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| Education | Student performance grouping |
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| Image Processing | Image segmentation |
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|             |                     |
|-------------|---------------------|
| Text Mining | Document clustering |
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|              |                     |
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| Social Media | Community detection |
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|            |                           |
|------------|---------------------------|
| E-commerce | Product/customer grouping |
|------------|---------------------------|

|               |                     |
|---------------|---------------------|
| Cybersecurity | Intrusion detection |
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|           |                       |
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| Geography | Region classification |
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### **5. Important Point**

The quality of clustering depends strongly on the **features selected** and the **similarity between data objects**.

Different clustering algorithms may produce different groups from the same dataset. Therefore, the selected algorithm should match the characteristics of the problem.

For example:

- **K-Means** → useful for relatively well-separated groups
- **Hierarchical Clustering** → useful when a hierarchy of groups is required
- **DBSCAN** → useful for irregularly shaped groups and noise
- **Fuzzy Clustering** → useful when an object may partially belong to multiple groups
- **BIRCH** → useful for large datasets
- **CURE** → useful for clusters of different shapes and sizes

## 6. Key Takeaway

**Clustering = Discovering natural groups in unlabeled data.**

It is widely used in **marketing, healthcare, finance, image processing, text mining, cybersecurity, social networks, recommendation systems, and geographical analysis.**

Before applying clustering, it is important to understand the **problem domain, available features, expected group structure, and appropriate clustering algorithm.**

